

SmartTAR STAR 1.3.0

Documentation

Windows TAR-based smart archiver with file-level deduplication, adaptive block compression and archive browsing.

SmartTAR is an MTB project - Make TAR Better

Version 1.3.0 | Project documentation | Format philosophy: TAR-based, file-as-atom, block-as-unit

Not RAR. Not 7z. Just TAR - made smarter for Windows.

1. Simple overview

SmartTAR is a Windows PowerShell GUI archiver built around the standard Windows tar.exe / bsdtar tool. It creates STAR archives: TAR-based containers that store data in independent TAR blocks and describe the archive through a JSON manifest.

The goal is not to replace TAR with a black-box format. The goal is to make TAR more practical for real-world folders, driver packs, installation media, old network shares and other messy datasets.

- STAR remains TAR-based. The outer archive is a PAX TAR container.
- Blocks are archive units. Each block is a .tar, .tar.xz or .tar.zst file.
- Files are dedup atoms. Identical files are stored once and restored through manifest aliases.
- The manifest is the map. It records the blocks, methods, hashes, dedup aliases and build summary.
- Browse mode lets the user inspect a STAR archive and extract selected files or folders.

SmartTAR is most useful when the input is not a single opaque blob, but a real file tree where duplicates, file types and compression behavior can be understood.

2. What SmartTAR can do

Area	Capability
Create STAR archives	Compresses files/folders into a STAR archive using Store, Balanced, Smart or Solid profiles.
Verify archives	Checks outer archive, internal blocks, block hashes and dedup alias targets.
Extract archives	Restores stored files and recreates dedup alias files from the manifest.
Browse archives	Shows a tree view of archived content and extracts selected file/folder.
Deduplicate files	Stores identical files once; all duplicates are restored using manifest aliases.
Adaptive compression	Chooses STORE, XZ9 or ZSTD19 per group; Smart mode probes archive-like data.
Large blocks	Uses PAX TAR for the outer STAR container, avoiding classic TAR entry size limits.

3. STAR archive structure

A STAR archive is an outer PAX TAR file with a predictable internal layout. The outer file normally has the .star extension, but it is still a TAR-family container.

Typical layout:

```
archive.star
├── blocks/
│   ├── 000001_structure.tar.xz
│   ├── 000002_stored.tar
│   ├── 000003_text.tar.xz
│   └── 000004_archives.tar.zst
└── manifest.json
```

Component	Purpose
Outer .star	PAX TAR container that holds all STAR entries.
blocks/	Folder with independent TAR block files.
structure block	Optional metadata-friendly block for directory structure.

Component	Purpose
data blocks	STORE, XZ9 or ZSTD19 TAR blocks containing the physical unique files.
manifest.json	Minified JSON map of the archive: blocks, hashes, dedup aliases and build summary.

4. Create pipeline

The create pipeline is designed to avoid false-success archives. A block is committed to the manifest only after it has been physically published into the outer STAR container and confirmed by listing the outer archive.

- Scan the selected source file or folder.
- Build a catalog and plan relative paths safely.
- Detect duplicate files and create manifest aliases for duplicates.
- Classify unique files into groups such as text, binary, archives, stored or compressible.
- Create block files using STORE, XZ9 or ZSTD19.
- Append each block to the outer STAR using PAX TAR.
- Confirm that the published block exists in the outer STAR before writing it to the manifest.
- Write minified manifest.json as the last outer entry.
- Optionally verify the final archive.

Reliability rule: SmartTAR does not trust only tar.exe exit codes. The program verifies the real state of the archive after critical publish operations.

5. Compression profiles

Profile	Behavior	Best use
Store	No content compression. Dedup still applies.	Fast baseline, already compressed data, diagnostics.
Balanced	Groups store-like data separately and compresses general compressible data with XZ9.	Best practical default for mixed file trees.
Smart	More detailed grouping; probes archives with STORE vs ZSTD19 vs XZ9.	Maximum practical compression without breaking TAR/block philosophy.
Solid	Single solid-like data grouping after dedup, method selected from profile.	Benchmark or maximum-context experiments.

6. Smart mode method probe

In version 1.3.0 the Smart profile no longer treats archive-like data as automatically uncompressible. The archives group is sampled and tested using STORE, ZSTD19 and XZ9. If a compressed method provides meaningful savings, the whole archives group is switched to the best method.

This keeps the logic adaptive: archive-like data remains STORE when compression is pointless, but becomes ZSTD19 or XZ9 when the actual content proves compressible.

7. File-level deduplication

SmartTAR deduplicates whole files. It does not split files into binary chunks and it does not use a custom content-addressed store. If two or more files have identical content, one physical file is stored in a block and all duplicate paths are recorded as dedup aliases in the manifest.

Principle	Description
File is atom	Dedup is performed on whole file content.
Alias restore	Duplicates are recreated during extraction by copying the stored target file.
Manifest-driven	Alias path, target path and byte count are stored in manifest.json.
Transparent recovery	Data blocks remain normal TAR files.

8. PAX TAR and publish-confirm

Classic TAR headers can fail with very large file entries. SmartTAR therefore uses POSIX PAX TAR for the outer .star container. Inner blocks remain normal TAR, TAR.XZ or TAR.ZST units.

The publish-confirm step protects the archive from a dangerous class of failures: tar.exe may print an error but still return exit code 0. SmartTAR therefore confirms that each block entry is actually present in the outer STAR before the block is added to the final manifest.

9. Manifest

The manifest is the archive map. It describes what each block contains, which compression method was used, where dedup aliases should be restored, and how the archive was built.

Version 1.3.0 writes the manifest as minified JSON using:

```
ConvertTo-Json -Depth 40 -Compress
```

The file remains plain JSON, only without unnecessary whitespace. This reduces manifest size while keeping compatibility simple.

10. Browse mode

The Add / Browse ARCHIVE button can now open a STAR archive in browse mode. The user chooses Add or Browse after selecting a .star file.

- Add selects the archive for normal Extract or Verify operations.
- Browse opens a tree view built from the archive blocks and manifest.
- The user can select a file or folder and extract only that item.
- Browse extraction respects the main Open output folder after success checkbox.

11. Reference benchmark from real-world test

The following numbers come from a real mounted installation image test. The source was a logical file tree with 71.03 GB of data, 30,083 files, 3,796 folders, 6,106 dedup aliases and 19.02 GB of alias bytes.

Profile	Archive size	Ratio	Saved	Verification
Store	23.69 GB	33.35 %	66.65 %	OK
Balanced	16.13 GB	22.71 %	77.29 %	OK

Profile	Archive size	Ratio	Saved	Verification
Smart before probe	16.31 GB	22.96 %	77.04 %	OK
Smart 1.3.0 probe	15.60 GB	21.96 %	78.04 %	OK

The probe result demonstrates why archive-like data should not always be forced to STORE. In that test, the archives group was switched to ZSTD19 and the Smart profile became the best result.

12. Design philosophy and limits

- STAR is intentionally TAR-based and transparent.
- Blocks remain independently readable TAR units.
- Dedup is file-level only; similar files are not delta-compressed.
- The manifest is required for full restore, especially for dedup aliases.
- Browse extraction currently prioritizes correctness over minimal I/O and may prepare blocks through a temporary payload.

13. Version 1.3.0 change summary

- Added Browse mode with tree view and selected item extraction.
- Changed Add ARCHIVE to Add / Browse ARCHIVE.
- Added two-button Add / Browse selection dialog.
- Fixed Browse extraction to respect Open output folder after success.
- Kept PAX outer TAR, publish-confirm, Smart probe and minified manifest from the final 1.2.2 branch.
- Updated UI and internal version to SmartTAR STAR 1.3.0.